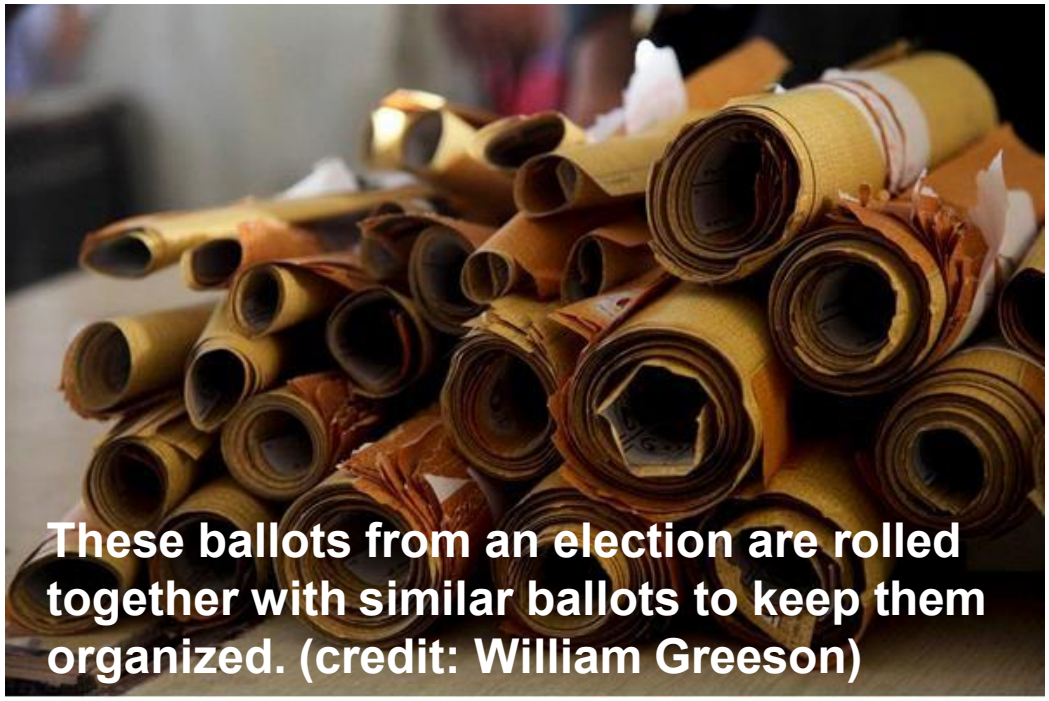
Chapter 2: DESCRIPTIVE STATISTICS

2.1 STEM-AND-LEAF GRAPHS (STEM PLOTS), LINE GRAPHS, AND BAR GRAPHS

Descriptive Statistics

In this chapter, you will study numerical and graphical ways to describe and display your data. This area of statistics is called “Descriptive Statistics.” You will learn how to calculate, and even more importantly, how to interpret these measurements and graphs.



These ballots from an election are rolled together with similar ballots to keep them organized. (credit: William Greeson)

Stem-and-Leaf Graphs

One simple graph, the stem-and-leaf graph or stem-plot, comes from the field of exploratory data analysis. It is a good choice when the data sets are small.

To create the plot, divide each observation of data into a stem and a leaf. The leaf consists of a final significant digit.

Stem	Leaf
3	3
4	2 9 9
5	3 5 5
6	1 3 7 8 8 9 9
7	2 3 4 8
8	0 3 8 8 8
9	0 2 4 4 4 4 6
10	0

Stem-and-Leaf Graphs

For example, 23 has stem two and leaf three. The number 432 has stem 43 and leaf two. Likewise, the number 5,432 has stem 543 and leaf two. The decimal 9.3 has stem nine and leaf three. Write the stems in a vertical line from smallest to largest. Draw a vertical line to the right of the stems. Then write the leaves in increasing order next to their corresponding stem.

Stem-and-Leaf Graphs

Example:

For Susan Dean's spring pre-calculus class, scores for the first exam were as follows (smallest to largest): 33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100

The stem-plot shows that most scores fell in the 60s, 70s, 80s, and 90s. Eight out of the 31 scores or approximately 26% were in the 90s or 100, a fairly high number of As.

Stem	Leaf
3	3
4	2 9 9
5	3 5 5
6	1 3 7 8 8 9 9
7	2 3 4 8
8	0 3 8 8 8
9	0 2 4 4 4 4 6
10	0

Example:

The data are the distances (in kilometers) from a home to local supermarkets. Create a stem-plot using the data: 1.1; 1.5; 2.3; 2.5; 2.7; 3.2; 3.3; 3.3; 3.5; 3.8; 4.0; 4.2; 4.5; 4.5; 4.7; 4.8; 5.5; 5.6; 6.5; 6.7; 12.3

Do the data seem to have any concentration of values?

The value 12.3 may be an outlier. Values appear to concentrate at three and four kilometers.

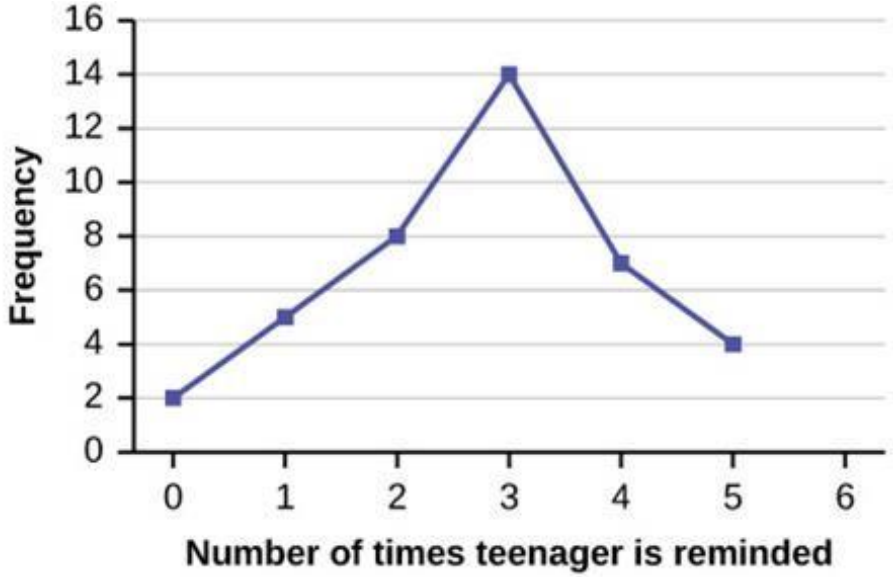
Stem	Leaf
1	1 5
2	3 5 7
3	2 3 3 5 8
4	0 2 5 5 7 8
5	5 6
6	5 7
7	
8	
9	
10	
11	
12	3

Line Graphs

Another type of graph that is useful for specific data values is a line graph.

Example

In a survey, 40 mothers were asked how many times per week a teenager must be reminded to do his or her chores. The results are shown in Table and in Figure.



Number of times teenager is reminded	Frequency
0	2
1	5
2	8
3	14
4	7
5	4

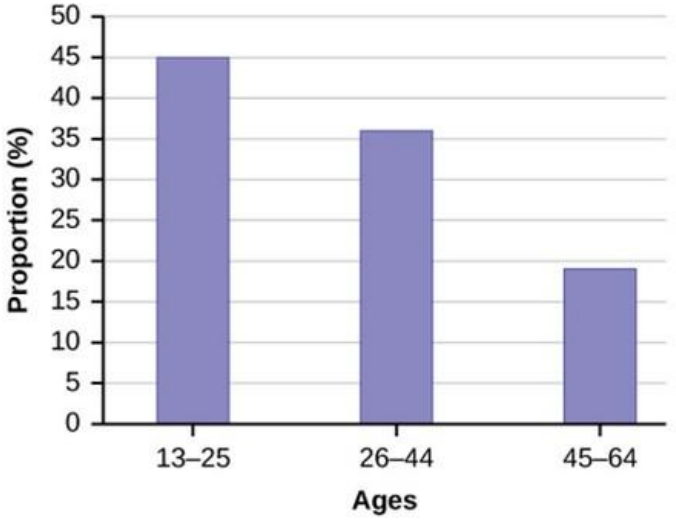
Bar graphs

Bar graphs consist of bars that are separated from each other. The bars can be rectangles or they can be rectangular boxes (used in three-dimensional plots), and they can be vertical or horizontal.

Example

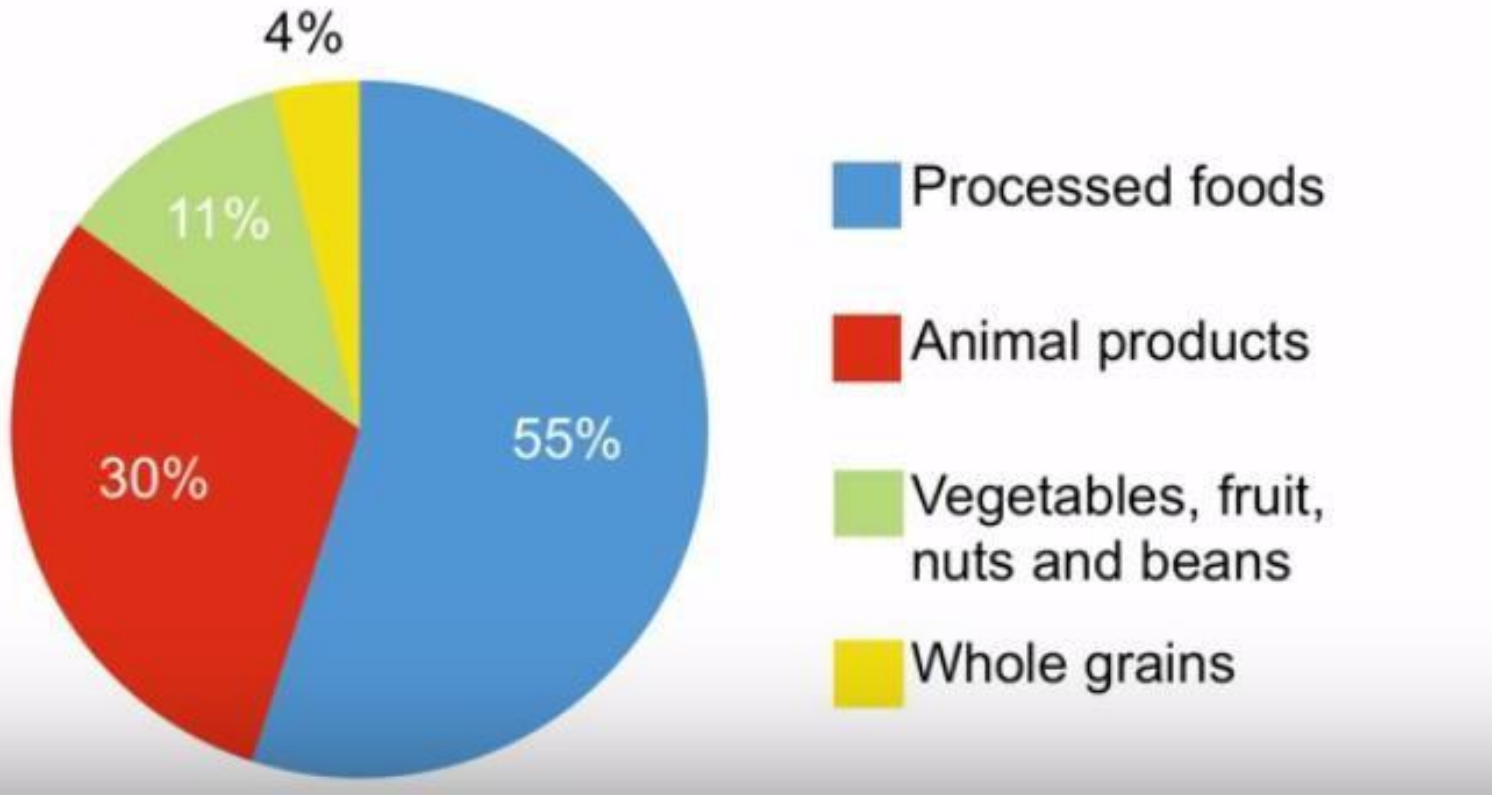
By the end of 2011, Facebook had over 146 million users in the United States. Table shows three age groups, the number of users in each age group, and the proportion (%) of users in each age group. Construct a bar graph using this data.

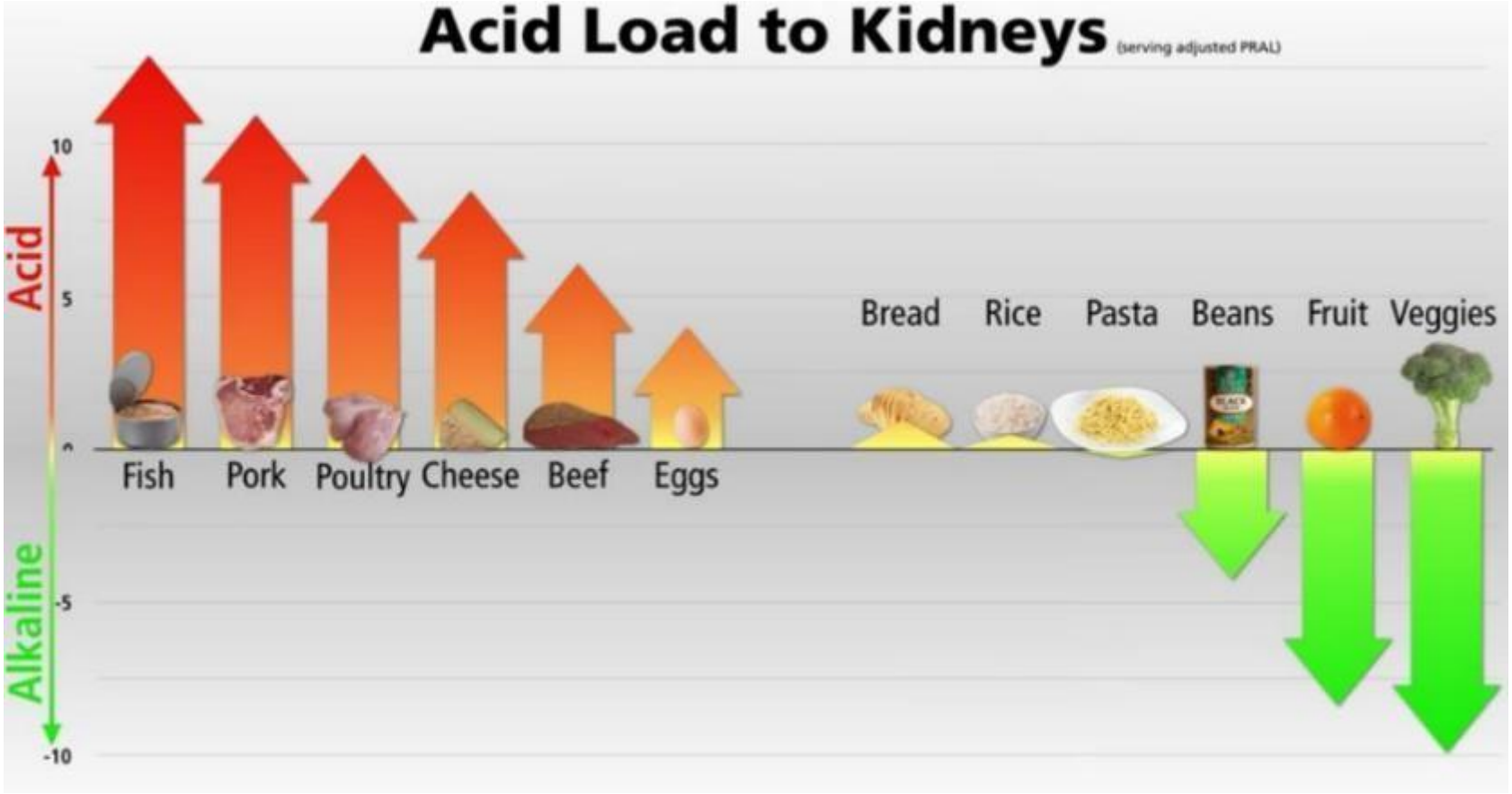
Age groups	Number of Facebook users	Proportion
13–25	65,082,280	45%
26–44	53,300,200	36%
45–64	27,885,100	19%

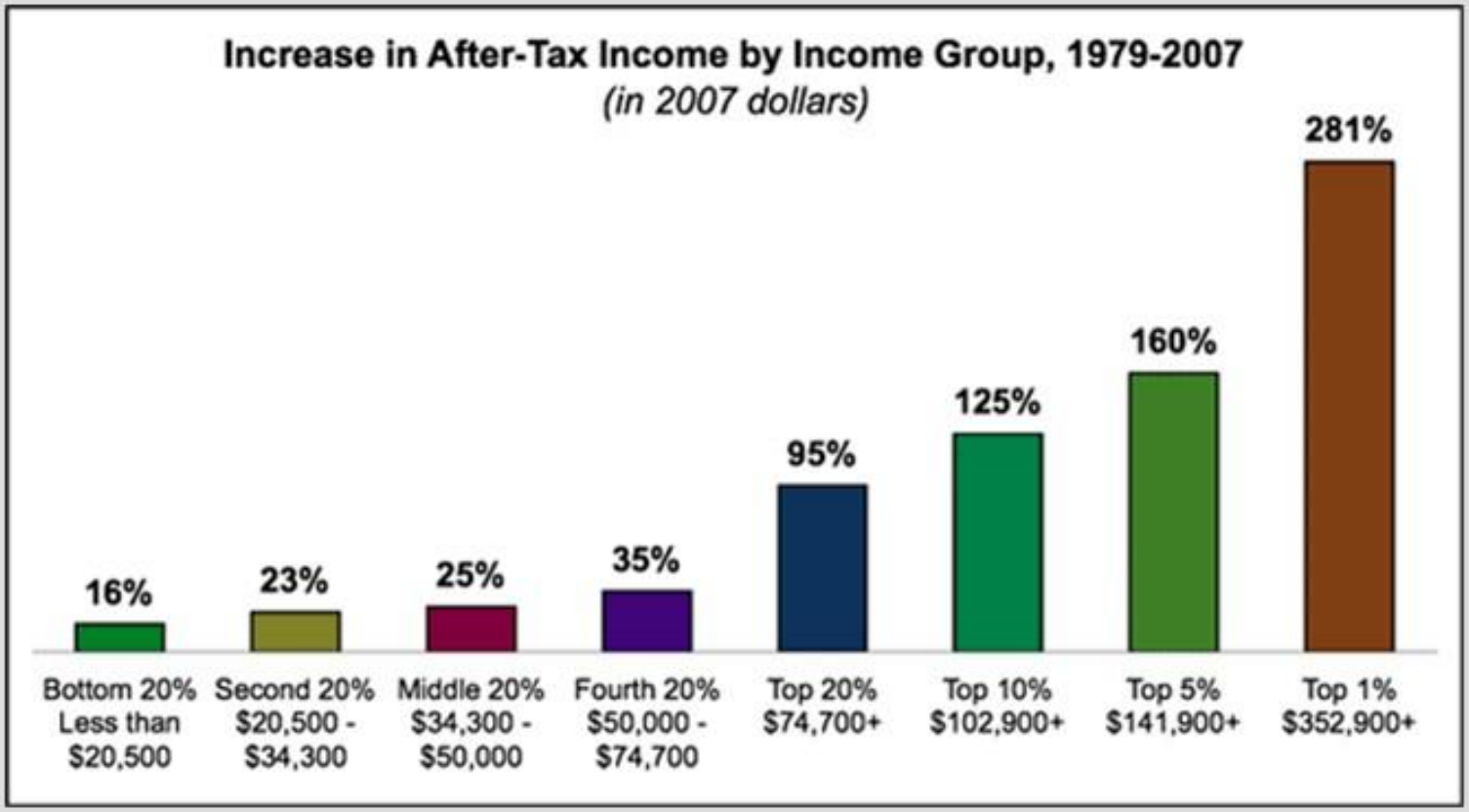


The Standard American Diet (SAD):

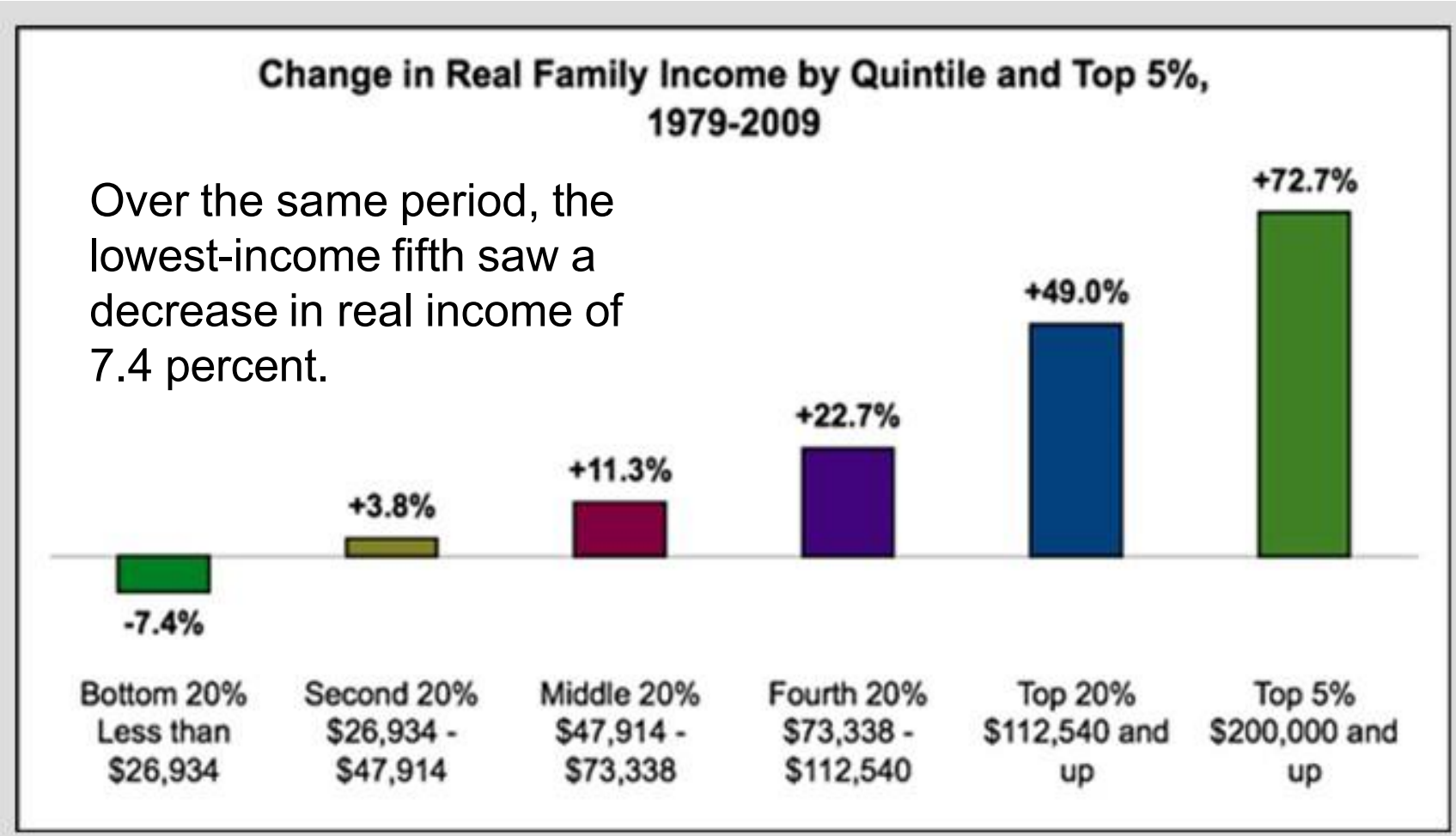
promotes chronic disease and weakens immune function




Acid Load to Kidneys
(serving adjusted PRAL)
Acid
Alkaline
10
5
0
-5
-10
Fish
Pork
Poultry
Cheese
Beef
Eggs
Bread
Rice
Pasta
Beans
Fruit
Veggies



Source: Congressional Budget Office, Average Federal Taxes by Income Group, "Average After-Tax Household Income," June, 2010.



Source: U.S. Census Bureau, Historical Income Tables: Families, Table F-3 (for income changes) and Table F-1 (for income ranges in 2009 dollars).

Between 1979 and 2009, the top 5 percent of American families saw their real incomes increase 72.7 percent, according to Census data.

2.2
HISTOGRAMS, FREQUENCY POLYGONS, AND TIME SERIES GRAPHS

Histogram

A histogram consists of contiguous (adjoining) boxes. It has both a horizontal axis and a vertical axis. The horizontal axis is labeled with what the data represents (for instance, distance from your home to school). The vertical axis is labeled either frequency or relative frequency (or percent frequency or probability). The graph will have the same shape with either label. The histogram (like the stem-plot) can give you the shape of the data, the center, and the spread of the data. A rule of thumb is to use a histogram when the data set consists of 100 values or more.

Histogram

To construct a histogram, first decide how many bars or intervals, also called classes, represent the data. Many histograms consist of five to 15 bars or classes for clarity. The number of bars needs to be chosen.

Choose a starting point for the first interval to be less than the smallest data value. A convenient starting point is a lower value carried out to one more decimal place than the value with the most decimal places.

For example, if the value with the most decimal places is 6.1 and this is the smallest value, a convenient starting point is 6.05 (6.1 – 0.05 = 6.05). We say that 6.05 has more precision.

If the value with the most decimal places is 2.23 and the lowest value is 1.5, a convenient starting point is 1.495 (1.5 – 0.005 = 1.495).

If the value with the most decimal places is 3.234 and the lowest value is 1.0, a convenient starting point is 0.9995 (1.0 – 0.0005 = 0.9995).

If all the data happen to be integers and the smallest value is two, then a convenient starting point is 1.5 (2 – 0.5 = 1.5).

Also, when the starting point and other boundaries are carried to one additional decimal place, no data value will fall on a boundary.

Example

The following data are the heights (in inches to the nearest half inch) of 100 male semiprofessional soccer players. The heights are continuous data, since height is measured.

60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

Example

The smallest data value is 60. Since the data with the most decimal places has one decimal (for instance, 61.5), we want our starting point to have two decimal places. Since the numbers 0.5, 0.05, 0.005, etc. are convenient numbers, use 0.05 and subtract it from 60, the smallest value, for the convenient starting point.

60 – 0.05 = 59.95 which is more precise than, say, 61.5 by one decimal place.

The starting point is, then, 59.95.

The largest value is 74, so 74 + 0.05 = 74.05 is the ending value.

60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

Example

Next, calculate the width of each bar or class interval. To calculate this width, subtract the starting point from the ending value and divide by the number of bars (you must choose the number of bars you desire). For example, if there are n values of data, take the square root of n and round to next integer. Suppose you choose eight bars.

$$\frac{74.05 - 59.95}{8} = 1.76$$

We will round up to two and make each bar or class interval two units wide.

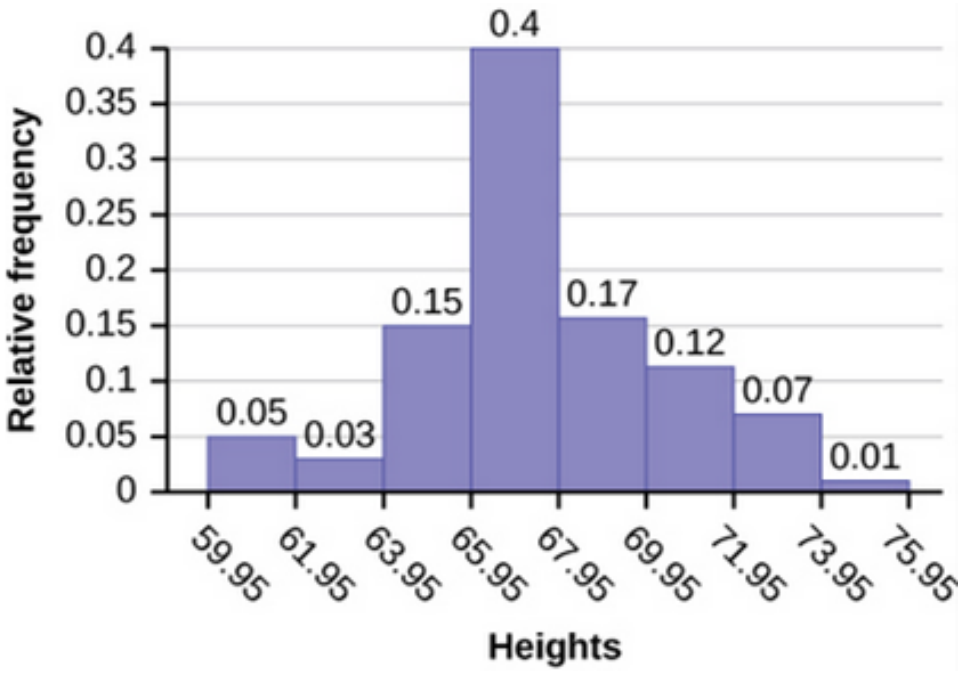
60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

Example

We will round up to two and make each bar or class interval two units wide.

The boundaries are:

- 59.95
- 59.95 + 2 = 61.95
- 61.95 + 2 = 63.95
- 63.95 + 2 = 65.95
- 65.95 + 2 = 67.95
- 67.95 + 2 = 69.95
- 69.95 + 2 = 71.95
- 71.95 + 2 = 73.95
- 73.95 + 2 = 75.95



60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

Frequency polygons

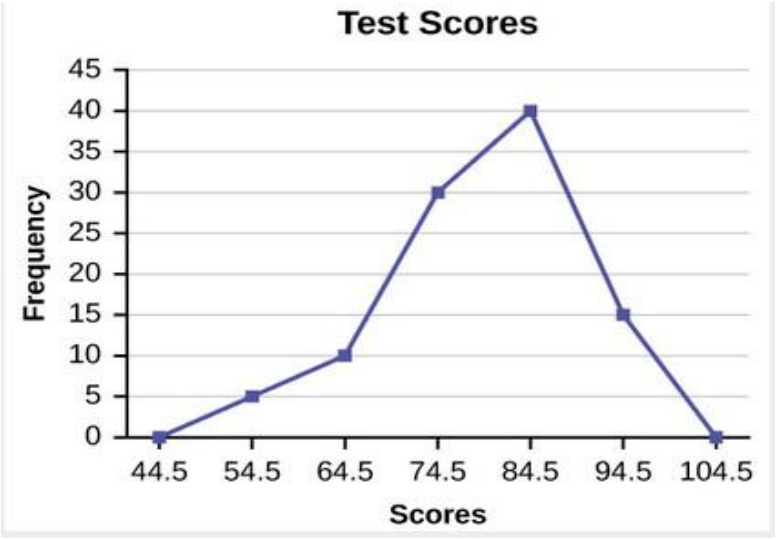
Frequency polygons are analogous to line graphs, and just as line graphs make continuous data visually easy to interpret, so too do frequency polygons.

To construct a frequency polygon, first organize the data as if you are graphing a histogram. Then, mark the midpoints of the intervals on the x-axis. Now, plot the (the midpoint, frequency) of each interval. Connect the points like you do with a line graph. Also, plot points one interval below the lowest midpoint with zero frequency, and one interval above the highest midpoint with zero frequency; note that these point are included to force the graph touch the x-axis to give it the closed polygon shape.

Example

Frequency Distribution for Calculus Final Test Scores:

Lower Bound	Upper Bound	Frequency	Cumulative Frequency
49.5	59.5	5	5
59.5	69.5	10	15
69.5	79.5	30	45
79.5	89.5	40	85
89.5	99.5	15	100



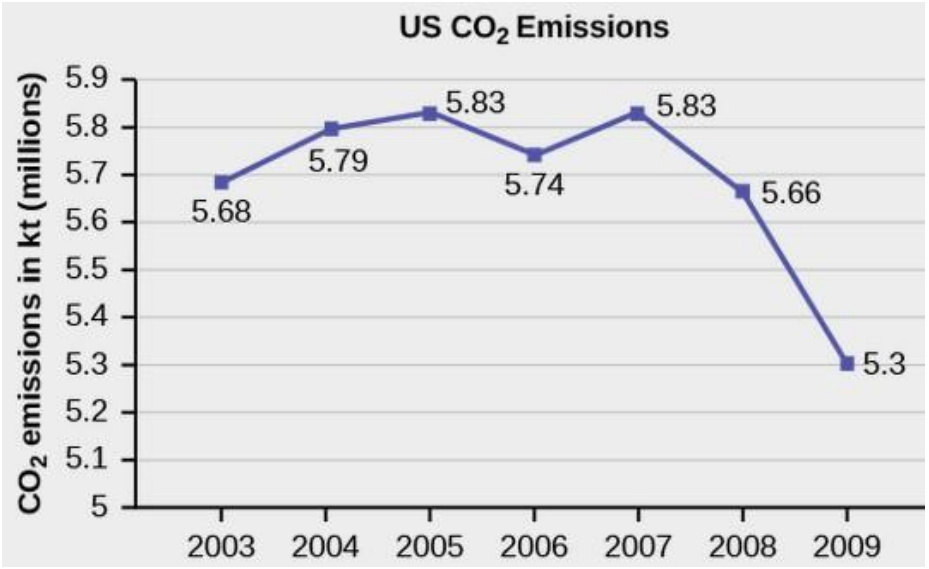
Time Series Graphs

To construct a time series graph, we must look at both pieces of our paired data set. We start with a standard Cartesian coordinate system. The horizontal axis is used to plot the date or time increments, and the vertical axis is used to plot the values of the variable that we are measuring. By doing this, we make each point on the graph correspond to a date and a measured quantity. The points on the graph are typically connected by straight lines in the order in which they occur.

Example

The following table is a portion of a data set from www.worldbank.org. Use the table to construct a time series graph for CO2 emissions for the United States.

United States	CO2 Emissions
2003	5,681,664
2004	5,790,761
2005	5,826,394
2006	5,737,615
2007	5,828,697
2008	5,656,839
2009	5,299,563



2.3
MEASURES OF THE LOCATION OF THE DATA

Percentile and Quartiles

To calculate quartiles and percentiles, the data must be ordered from smallest to largest. Quartiles divide ordered data into quarters. Percentiles divide ordered data into hundredths. To score in the 90th percentile of an exam does not mean, necessarily, that you received 90% on a test. It means that 90% of test scores are the same or less than your score and 10% of the test scores are the same or greater than your test score.

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Formulas for Finding the kth Percentile and kth Quartile:

$$i = \frac{k}{100}(n+1)$$

$$i = \frac{k}{4}(n+1)$$

Example

Listed are 29 ages for Academy Award winning best actors in order from smallest to largest.

18; 21; 22; 25; 26; 27; 29; 30; 31; 33; 36; 37; 41; 42; 47; 52; 55; 57; 58; 62; 64; 67; 69; 71; 72; 73; 74; 76; 77

Find the 70th percentile.
Find the 83rd percentile.

$$i = \frac{k}{100}(n + 1) = \frac{70}{100}(29+1) = 21.$$

Twenty-one is an integer, and the data value in the 21st position in the ordered data set is 64. The 70th percentile is 64 years.

$$i = \frac{k}{100}(n+1) = \frac{83}{100}(29+1) = 24.9.$$

24.9 is NOT an integer. Round it down to 24 and up to 25. The age in the 24th position is 71 and the age in the 25th position is 72. Average 71 and 72. The 83rd percentile is 71.5 years.

Example

For the following 13 real estate prices, calculate the IQR and determine if any prices are potential outliers. Prices are in dollars.

389,950; 230,500; 158,000; 479,000; 639,000; 114,950; 5,500,000; 387,000; 659,000; 529,000; 575,000; 488,800; 1,095,000

Order the data from smallest to largest. The IQR is the distance between Q1 and Q3.

114,950; 158,000; 230,500; 387,000; 389,950; 479,000; 488,800; 529,000; 575,000; 639,000; 659,000; 1,095,000; 5,500,000

$$i = \frac{k}{4}(n + 1) = \frac{1}{4}(13+1) = 3.5,$$

so, round down and up, and take the average of 3rd and 4th:

$$Q_1 = \frac{230,500 + 387,000}{2} = 308,750$$

$$i = \frac{k}{4}(n + 1) = \frac{3}{4}(13+1) = 10.5,$$

so, round down and up, and take the average of 10th and 11th:

$$Q_3 = \frac{639,000 + 659,000}{2} = 649,000$$

Order the data from smallest to largest. The IQR is the distance between Q1 and Q3.

114,950; 158,000; 230,500; 387,000; 389,950; 479,000; 488,800; 529,000; 575,000; 639,000; 659,000; 1,095,000; 5,500,000

$$i = \frac{k}{4}(n + 1) = \frac{1}{4}(13+1) = 3.5,$$

so, round down and up, and take the average of 3rd and 4th:

$$Q_1 = \frac{230,500 + 387,000}{2} = 308,750$$

$$i = \frac{k}{4}(n + 1) = \frac{3}{4}(13+1) = 10.5,$$

so, round down and up, and take the average of 10th and 11th:

$$Q_3 = \frac{639,000 + 659,000}{2} = 649,000$$

IQR = 649,000 – 308,750 = 340,250

(1.5)(IQR) = (1.5)(340,250) = 510,375

Q_1 – (1.5)(IQR) = 308,750 – 510,375 = –201,625
Q_3 + (1.5)(IQR) = 649,000 + 510,375 = 1,159,375

No house price is less than –201,625. However, 5,500,000 is more than 1,159,375. Therefore, 5,500,000 is a potential outlier.

A Formula for Finding the Percentile of a Value in a Data Set:
Order the data from smallest to largest.

Calculate $\frac{x+0.5y}{n}(100)$, then round to the nearest integer.

Where:
x = the number of data values counting from the bottom of the data list up to but not including the data value for which you want to find the percentile.

y = the number of data values equal to the data value for which you want to find the percentile.

n = the total number of data.

Example

Listed are 29 ages for Academy Award winning best actors in order from smallest to largest.

18; 21; 22; 25; 26; 27; 29; 30; 31; 33; 36; 37; 41; 42; 47; 52; 55; 57; 58; 62; 64; 67; 69; 71; 72; 73; 74; 76; 77

Find the percentile for 58.

Find the percentile for 25.

Counting from the bottom of the list, there are 18 *data values less than* 58. *There is one value of* 58, *so for* $x = 18$ *and* $y = 1$:.

$$\frac{x + 0.5y}{n}(100) = \frac{18 + 0.5(1)}{29}(100) = 63.8.$$

So, 58 *is the* 64*th percentile.*

Counting from the bottom of the list, there are 3 *data values less than* 25. *There is one value of* 25, *so for* $x = 3$ *and* $y = 1$:.

$$\frac{x + 0.5y}{n}(100) = \frac{3 + 0.5(1)}{29}(100) = 12.07.$$

So, 25 *is the* 12*th percentile.*

Interpreting Percentiles, Quartiles, and Median

Median is the 50th Percentile or the 2nd Quartile. A percentile may or may not correspond to a value judgment about whether it is “good” or “bad.” The interpretation of whether a certain percentile is “good” or “bad” depends on the context of the situation to which the data applies. In some situations, a low percentile would be considered “good;” in other contexts a high percentile might be considered "good". In many situations, there is no value judgment that applies.

Example

On a timed math test, the first quartile for time it took to finish the exam was 35 minutes. Interpret the first quartile in the context of this situation.

- Twenty-five percent of students finished the exam in 35 minutes or less.
- Seventy-five percent of students finished the exam in 35 minutes or more.
- A low percentile could be considered good, as finishing more quickly on a timed exam is desirable. (If you take too long, you might not be able to finish.)

Example

At a community college, it was found that the 30th percentile of credit units that students are enrolled for is seven units. Interpret the 30th percentile in the context of this situation.

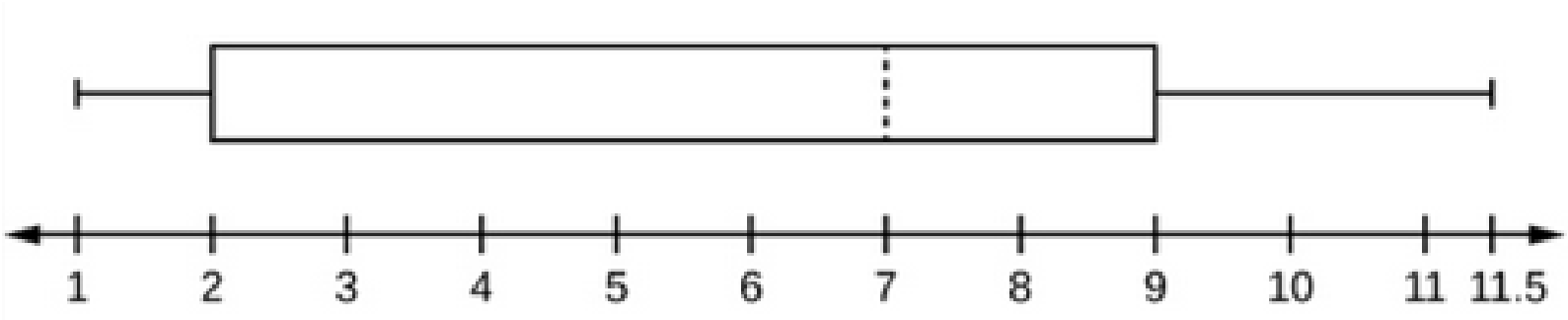
- Thirty percent of students are enrolled in seven or fewer credit units.
- Seventy percent of students are enrolled in seven or more credit units.
- In this example, there is no “good” or “bad” value judgment associated with a higher or lower percentile. Students attend community college for varied reasons and needs, and their course load varies according to their needs.

2.4
BOX PLOTS

Box plots

Box plots (also called box-and-whisker plots or box-whisker plots) give a good graphical image of the concentration of the data. They also show how far the extreme values are from most of the data.

A box plot is constructed from five values: the minimum value, the first quartile, the median, the third quartile, and the maximum value. We use these values to compare how close other data values are to them.



Box plots

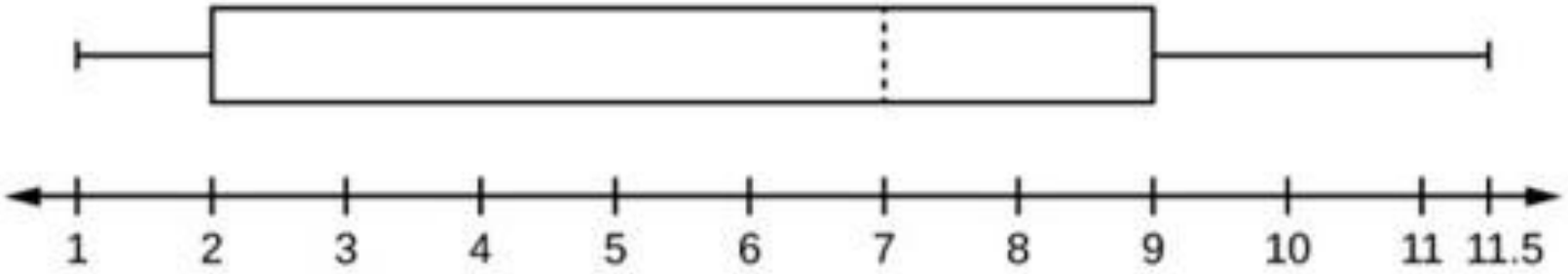
Approximately the middle 50 percent of the data fall inside the box. The “whiskers” extend from the ends of the box to the smallest and largest data values. The median or second quartile can be between the first and third quartiles, or it can be one, or the other, or both. The box plot gives a good, quick picture of the data.

Example

Consider, again, this dataset.

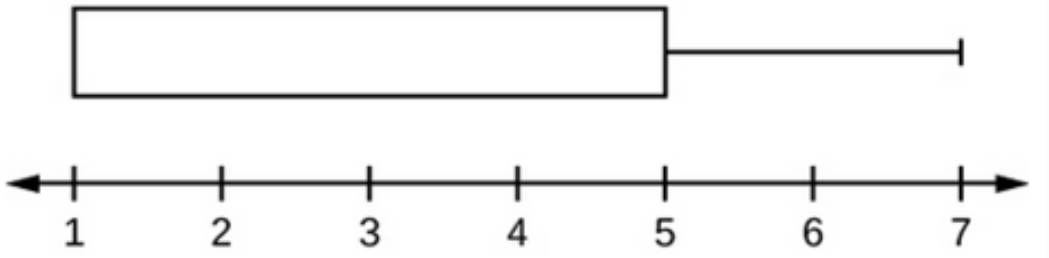
1 1 2 2 4 6 6.8 7.2 8 8.3 9 10 10 11.5

The first quartile is two, the median is seven, and the third quartile is nine. The smallest value is one, and the largest value is 11.5. The following image shows the constructed box plot.



Box plots

For some sets of data, some of the largest value, smallest value, first quartile, median, and third quartile may be the same. For instance, you might have a data set in which the median and the third quartile are the same. In this case, the diagram would not have a dotted line inside the box displaying the median. The right side of the box would display both the third quartile and the median. For example, if the smallest value and the first quartile were both one, the median and the third quartile were both five, and the largest value was seven, the box plot would look like:



In this case, at least 25% of the values are equal to one. Twenty-five percent of the values are between one and five, inclusive. At least 25% of the values are equal to five. The top 25% of the values fall between five and seven, inclusive.

2.5
MEASURES OF THE CENTER OF THE DATA

The Mean and Median

The “center” of a data set is also a way of describing location. The two most widely used measures of the “center” of the data are the mean (average) and the median. To calculate the mean weight of 50 people, add the 50 weights together and divide by 50. To find the median weight of the 50 people, order the data and find the number that splits the data into two equal parts. The median is generally a better measure of the center when there are extreme values or outliers because it is not affected by the precise numerical values of the outliers. The mean is the most common measure of the center.

A Little Math Review

To understand the formulas in this course you need to know about the sigma notation. In the next few slides we review the concept.

Sigma Notation

Given a sequence

$$a_1, a_2, a_3, a_4, \ldots$$

we can write the sum of the first n terms using **summation notation**, or **sigma notation**. This notation derives its name from the Greek letter Σ(capital sigma, corresponding to our *S* for “sum”). Sigma notation is used as follows:

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + a_4 + \cdots + a_n$$

Sigma Notation

$$\sum_{k=1}^{m}(k+1)^2 = (1+1)^2 + (2+1)^2 + \cdots + (m+1)^2$$

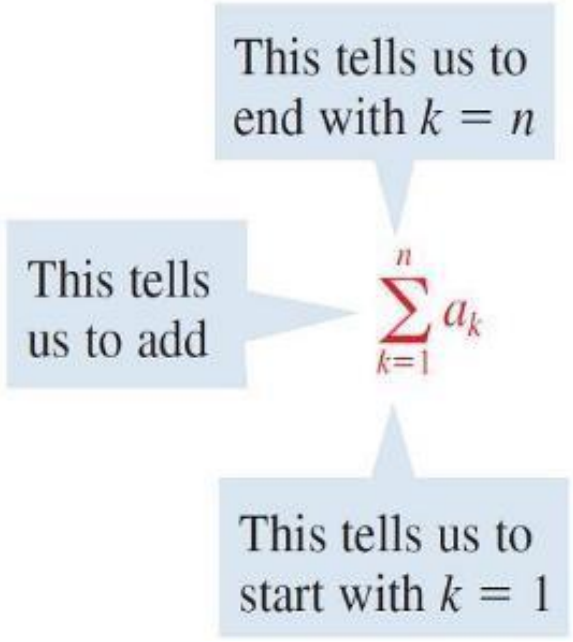
$$\sum_{i=1}^{m} f_i = f_1 + f_2 + f_3 + \cdots + f_m$$

$$\sum_{i=1}^{m} \frac{f_i}{n} = \frac{f_1}{n} + \frac{f_2}{n} + \frac{f_3}{n} + \cdots + \frac{f_m}{n}$$

Sigma Notation

The left side of this expression is read, “The sum of a_k from $k = 1$ to $k = n$.”

The letter k is called the **index of summation**, or the **summation variable**, and the idea is to replace k in the expression after the sigma by the integers 1, 2, 3, . . . , n, and add the resulting expressions, arriving at the right side of the equation.



Find each sum.

(a) $\sum_{k=1}^{5} k^2$ (b) $\sum_{j=3}^{5} \frac{1}{j}$ (c) $\sum_{i=5}^{10} i$ (d) $\sum_{i=1}^{6} 2$

Solution:

(a) $\sum_{k=1}^{5} k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

(b) $\sum_{j=3}^{5} \frac{1}{j} = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{60}$

(c) $\sum_{i=5}^{10} i = 5 + 6 + 7 + 8 + 9 + 10 = 45$

(d) $\sum_{i=1}^{6} 2 = 2 + 2 + 2 + 2 + 2 + 2 = 12$

Formulas to Find the Mean

To find the mean we add the values of a variable, and divide by the number of values. The letter used to represent the sample mean is an x with a bar over it (pronounced “x bar”) and the sample size is n:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

The Greek letter μ (pronounced “mew”) represents the population mean, and the population size is N. One of the requirements for the sample mean to be a good estimate of the population mean is for the sample taken to be truly random.

Example

Find the mean, consider the sample:

1; 1; 1; 2; 2; 3; 4; 4; 4; 4; 4

$$\bar{x} = \frac{1+1+1+2+2+3+4+4+4+4+4}{11} = 2.7$$

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

When each value in the data set is not unique, the mean can be calculated by multiplying each distinct value by its frequency and then dividing the sum by the total number of data values.

$$\bar{x} = \frac{3(1)+2(2)+1(3)+5(4)}{11} = 2.7$$

$$\overline{x} = \frac{\sum_{i=1}^{m} x_i f_i}{\sum_{i=1}^{m} f_i}, \text{ where } \sum_{i=1}^{m} f_i = n$$

x_i	f_i
1	3
2	2
3	1
4	5

A Formula to Find the Median

You can quickly find the location of the median by using the expression below, which is the simplified form of the formula

To find the 50th percentile, the Median.

$$i = \frac{k}{100}(n + 1) \xrightarrow{k=50} \frac{50}{100}(n+1) = \frac{n+1}{2}.$$

Example

AIDS data indicating the number of months a patient with AIDS lives after taking a new antibody drug are as follows (smallest to largest):

3; 4; 8; 8; 10; 11; 12; 13; 14; 15; 15; 16; 16; 17; 17; 18; 21; 22; 22; 24; 24; 25; 26; 26; 27; 27; 29; 29; 31; 32; 33; 33; 34; 34; 35; 37; 40; 44; 44; 47;

Calculate the mean and the median.

Example

AIDS data indicating the number of months a patient with AIDS lives after taking a new antibody drug are as follows (smallest to largest):

3; 4; 8; 8; 10; 11; 12; 13; 14; 15; 15; 16; 16; 17; 17; 18; 21; 22; 22; 24; 24; 25; 26; 26; 27; 27; 29; 29; 31; 32; 33; 33; 34; 34; 35; 37; 40; 44; 44; 47;

Calculate the mean and the median.

$$i = \frac{k}{100}(n + 1) = \frac{50}{100}(n+1) = \frac{n+1}{2}.$$

The calculation for the mean is:

$$\bar{x} = \frac{[3+4+(8)(2)+10+11+12+13+14+(15)(2)+(16)(2)+...+35+37+40+(44)(2)+47]}{40} = 23.6$$

To find the median, *M*, first use the formula for the location. The location is:

$$\frac{n+1}{2} = \frac{40+1}{2} = 20.5$$

Starting at the smallest value, the median is located between the 20th and 21st values (the two 24s):

3; 4; 8; 8; 10; 11; 12; 13; 14; 15; 15; 16; 16; 17; 17; 18; 21; 22; 22; 24; 24; 25; 26; 26; 27; 27; 29; 29; 31; 32; 33; 33; 34; 34; 35; 37; 40; 44; 44; 47;

$$M = \frac{24+24}{2} = 24$$

The Mode

Another measure of the center is the mode. The mode is the most frequent value. There can be more than one mode in a data set as long as those values have the same frequency and that frequency is the highest. A data set with two modes is called bimodal.

Example

Statistics exam scores for 20 students are as follows:

50 53 59 59 63 63 72 72 72 72 72 76 78 81 83 84 84 84 90 93

Find the mode.

The most frequent score is 72, which occurs five times.
Mode = 72.

Calculating the Mean of Grouped Frequency Tables

$$\bar{x} = \frac{\sum_{i=1}^{m} x_i f_i}{\sum_{i=1}^{m} f_i}, \text{ where } \sum_{i=1}^{m} f_i = n$$

Example

A frequency table displaying professor Blount's last statistic test is shown. Find the best estimate of the class mean.

Grade Interval	Number of Students
50–56.5	1
56.5–62.5	0
62.5–68.5	4
68.5–74.5	4
74.5–80.5	2
80.5–86.5	3
86.5–92.5	4
92.5–98.5	1

2.6
SKEWNESS AND THE MEAN, MEDIAN, AND MODE

$$\text{Kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4$$

$$skewness = \frac{(x - \mu)^3}{\sigma^3}$$

$$kurtosis = \frac{(x - \mu)^4}{\sigma^4}$$

Moment number	Name	Measure of	Formula
1	Mean	Central tendency	$\bar{X} = \frac{\sum_{i=1}^{N} X_i}{N}$
2	Variance (Volatility)	Dispersion	$\sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \bar{X})^2}{N}$
3	Skewness	Symmetry (Positive or Negative)	$Skew = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(X_i - \bar{X})}{\sigma} \right]^3$
4	Kurtosis	Shape (Tall or flat)	$Kurt = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(X_i - \bar{X})}{\sigma} \right]^4$

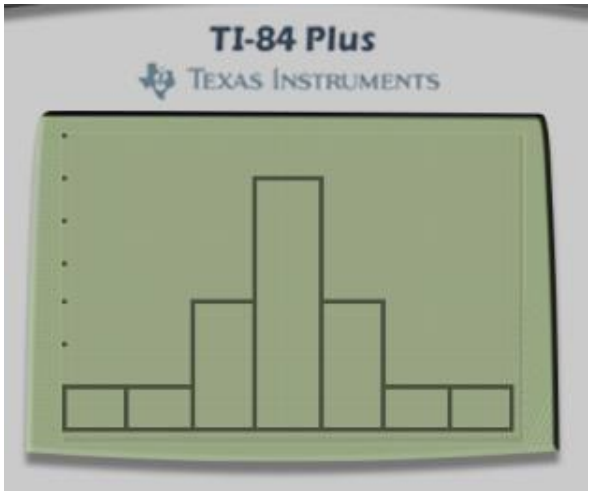
Where X is a random variable having N observations (i = 1,2,…,N).

Skewness and the Mean, Median, and Mode

Consider the following data set.

4; 5; 6; 6; 6; 7; 7; 7; 7; 7; 7; 8; 8; 8; 9; 10

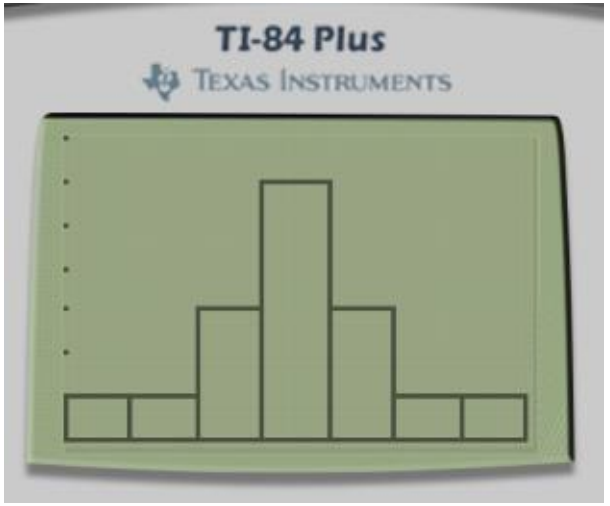
The histogram displays a symmetrical distribution of data. A distribution is symmetrical if a vertical line can be drawn at some point in the histogram such that the shape to the left and the right of the vertical line are mirror images of each other. The mean, the median, and the mode are each seven for these data. In a perfectly symmetrical distribution, the mean and the median are the same. This example has one mode (unimodal), and the mode is the same as the mean and median. In a symmetrical distribution that has two modes (bimodal), the two modes would be different from the mean and median.



Skewness and the Mean, Median, and Mode

Consider the following data set.

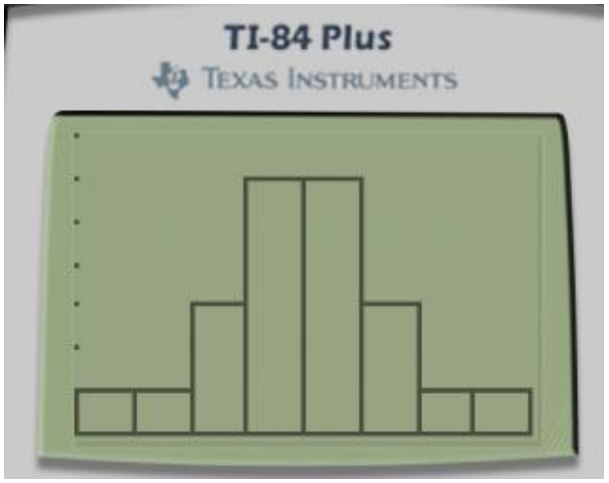
4; 5; 6; 6; 6; 7; 7; 7; 7; 7; 7; 8; 8; 8; 9; 10



In a symmetrical distribution that has two modes (bimodal), the two modes would be different from the mean and median. Compare these sets and their Histograms.

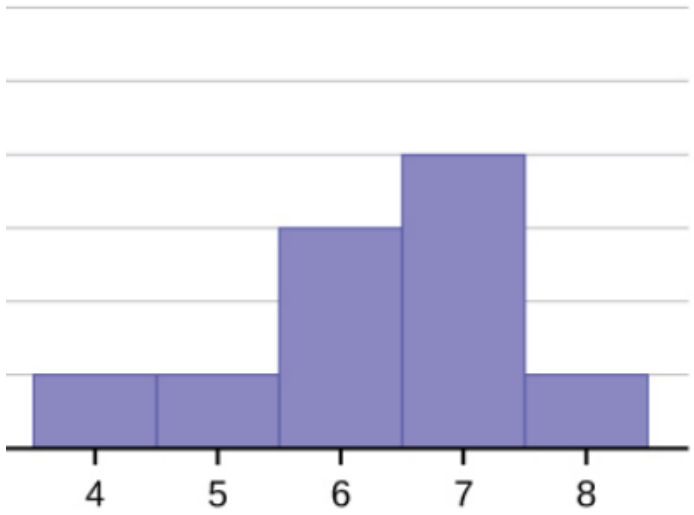
Consider the following data set.

4; 5; 6; 6; 6; 7; 7; 7; 7; 7; 7; 8; 8; 8; 8; 8; 8; 9; 9; 10; 11



Skewness and the Mean, Median, and Mode

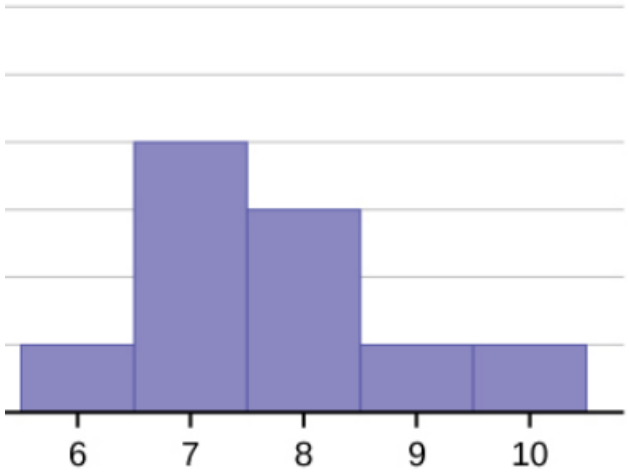
The histogram for the data: 4; 5; 6; 6; 6; 7; 7; 7; 7; 8 is not symmetrical. The right-hand side seems “chopped off” compared to the left side. A distribution of this type is called skewed to the left because it is pulled out to the left.



The mean is 6.3, the median is 6.5, and the mode is seven. Notice that the mean is less than the median, and they are both less than the mode. The mean and the median both reflect the skewing, but the mean reflects it more so.

Skewness and the Mean, Median, and Mode

The histogram for the data: 6; 7; 7; 7; 7; 8; 8; 8; 9; 10, is also not symmetrical. It is skewed to the right.



The mean is 7.7, the median is 7.5, and the mode is seven. Of the three statistics, the mean is the largest, while the mode is the smallest. Again, the mean reflects the skewing the most.

To summarize, generally if the distribution of data is skewed to the left, the mean is less than the median, which is often less than the mode. If the distribution of data is skewed to the right, the mode is often less than the median, which is less than the mean.

2.7
MEASURES OF THE SPREAD OF THE DATA

The standard deviation and variance

Your concentration should be on what the deviation tells us about the data. Here is an example to help us understand what the standard deviation is, and what it tells us about the data set.

Example

Here is the list of ages of all close friends of Jasmine. The mean of the ages is 41.5.

41 38 39 45 47 41 44 41 37 42

Every friend is either older than the average or younger. In other words, the ages deviate from the mean age. The question is: is there a number we can calculate that gives a measure for the overall deviation from the mean? Yes!

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

The table below lists the friends with coded names. A good start is to calculate the deviations from the mean, and add them.

Ages	Deviation: $x - \mu$
A=41	41 – 41.5 = –0.5
B=38	38 – 41.5 = –3.5
C=39	39 – 41.5 = –2.5
D=45	45 – 41.5 = 3.5
E=47	47 – 41.5 = 5.5
F=41	41 – 41.5 = –0.5
G=44	44 – 41.5 = 2.5
H=41	41 – 41.5 = –0.5
L=37	37 – 41.5 = –4.5
M=42	42 – 41.5 = 0.5

The problem is that the sum will always be zero!

$$\sum_{i=1}^{N}(x_i - \mu) = 0$$

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

The remedy is to square the deviations, or take absolute value of them to make them all positive. We'll square the deviations and add them.

Squares: $(x - \mu)^{2b}$
$(-0.5)^2 = 0.25$
$(-3.5)^2 = 12.25$
$(-2.5)^2 = 6.25$
$(3.5)^2 = 12.25$
$(5.5)^2 = 30.25$
$(-0.5)^2 = 0.25$
$(2.5)^2 = 6.25$
$(-0.5)^2 = 0.25$
$(-4.5)^2 = 20.25$
$(0.5)^2 = 0.25$

The problem is solved! Now, what if we divide this number by the total number of close friends? Well, we call the result the variance of the ages.

$$\sum_{i=1}^{N}(x_i - \mu)^2 = 88.5$$

$$\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} = \frac{88.5}{10} = 8.85$$

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

8.85 is an overall measure of square ages of Juan's friends, called the variance. But, square ages doesn't make any sense in the context of the problem; can we remedy this? Yes! Just take square root of this number. The result is called the standard deviation of the ages, 2.97.

Squares: $(x - \mu)^{2b}$
$(-0.5)^2 = 0.25$
$(-3.5)^2 = 12.25$
$(-2.5)^2 = 6.25$
$(3.5)^2 = 12.25$
$(5.5)^2 = 30.25$
$(-0.5)^2 = 0.25$
$(2.5)^2 = 6.25$
$(-0.5)^2 = 0.25$
$(-4.5)^2 = 20.25$
$(0.5)^2 = 0.25$

Take square root of the variance.

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} = \frac{88.5}{10} = 8.85$$

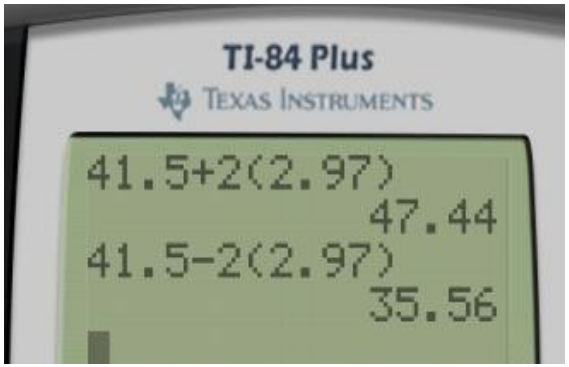
$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}} = \sqrt{8.85} = 2.97$$

The standard deviation and variance

A data value that is two standard deviations from the average is just on the borderline for what many statisticians would consider to be far from the average. Considering data to be far from the mean if it is more than two standard deviations away is more of an approximate “rule of thumb” than a rigid rule. In general, the shape of the distribution of the data affects how much of the data is further away than two standard deviations.

Note that all ages of close friends of Jasmine are within 2 standard deviations Of the mean, between 35.56 and 47.44.

41 38 39 45 47 41 44 41 37 42



The standard deviation and variance

If the numbers come from a census of the entire population and not a sample, when we calculate the average of the squared deviations to find the variance, we divide by N, the number of items in the population. If the data are from a sample rather than a population, when we calculate the average of the squared deviations, we divide by n – 1, one less than the number of items in the sample.

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x-\bar{x})^2}{n-1}}, \quad s = \sqrt{\frac{\sum_{i=1}^{m}(x-\bar{x})^2 f_i}{n-1}}, \quad where \quad n = \sum_{i=1}^{m} f_i$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x-\mu)^2}{N}}, \quad \sigma = \sqrt{\frac{\sum_{i=1}^{m}(x-\mu)^2 f_i}{N}}, \quad N = \sum_{i=1}^{m} f_i$$

The standard deviation and variance

For the sample variance, we divide by the sample size minus one (n – 1). Why not divide by n? The answer has to do with the population variance. The sample variance is an estimate of the population variance. Based on the theoretical mathematics that lies behind these calculations, dividing by (n – 1) gives a better estimate of the population variance.

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x-\bar{x})^2}{n-1}}, \quad s = \sqrt{\frac{\sum_{i=1}^{m}(x-\bar{x})^2 f_i}{n-1}}, \quad where \quad n = \sum_{i=1}^{m} f_i$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x-\mu)^2}{N}}, \quad \sigma = \sqrt{\frac{\sum_{i=1}^{m}(x-\mu)^2 f_i}{N}}, \quad N = \sum_{i=1}^{m} f_i$$

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

Now, if Jasmine had thousands of social media friends, and the list of ages were a sample of her friends, then we would have to slightly modify our calculation of the variance and standard deviation.

Squares: $(x - \mu)^{2b}$
$(-0.5)^2 = 0.25$
$(-3.5)^2 = 12.25$
$(-2.5)^2 = 6.25$
$(3.5)^2 = 12.25$
$(5.5)^2 = 30.25$
$(-0.5)^2 = 0.25$
$(2.5)^2 = 6.25$
$(-0.5)^2 = 0.25$
$(-4.5)^2 = 20.25$
$(0.5)^2 = 0.25$

$$s^2 = \frac{\sum_{i=1}^{n}(x - \bar{x})^2}{n-1} = \frac{88.5}{9} = 9.83$$

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x - \bar{x})^2}{n-1}}, \quad s = \sqrt{\frac{88.5}{9}} = 3.14$$

Example

Use the following data (first exam scores) from Susan Dean's spring pre-calculus class:

33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100

Create a chart containing the data, frequencies, relative frequencies, and cumulative relative frequencies to three decimal places.

Calculate the following to one decimal place using a TI-83+ or TI-84 calculator:

- The sample mean
- The sample standard deviation
- The median
- The first quartile
- The third quartile
- IQR

Construct a box plot and a histogram on the same set of axes. Make comments about the box plot, the histogram, and the chart.

Example

Use the following data (first exam scores) from Susan Dean's spring pre-calculus class:

33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100

Create a chart containing the data, frequencies, relative frequencies, and cumulative relative frequencies to three decimal places.

Data	Frequency	Relative Frequency	Cumulative Relative Frequency
33	1	0.032	0.032
42	1	0.032	0.064
49	2	0.065	0.129
53	1	0.032	0.161
55	2	0.065	0.226
61	1	0.032	0.258
63	1	0.032	0.29
67	1	0.032	0.322
68	2	0.065	0.387
69	2	0.065	0.452
72	1	0.032	0.484
73	1	0.032	0.516
74	1	0.032	0.548
78	1	0.032	0.580
80	1	0.032	0.612
83	1	0.032	0.644
88	3	0.097	0.741
90	1	0.032	0.773
92	1	0.032	0.805
94	4	0.129	0.934
96	1	0.032	0.966
100	1	0.032	0.998

Note that relative frequencies don't add to 1 because of rounding errors.

Comparing Values from Different Data Sets

The standard deviation is useful when comparing data values that come from different data sets. If the data sets have different means and standard deviations, then comparing the data values directly can be misleading. In such cases we calculate the z score for each value.

For each data value, calculate how many standard deviations away from its mean the value is. Find the distance between the value and the mean, then divide by standard deviation to find how many standard deviations fit in the distance. Use the formulas below for values from a population or a sample.

$$For\ values\ from\ a\ sample: \quad z = \frac{x - \bar{x}}{s}$$

$$For\ values\ from\ a\ population: \quad z = \frac{x - \mu}{\sigma}$$

Comparing Values from Different Data Sets

Two students, John and Ali, from different high schools, wanted to find out who had the highest GPA when compared to his school. Which student had the highest GPA when compared to his school?

Student	GPA	School Mean GPA	School Standard Deviation
John	2.85	3.0	0.7
Ali	77	80	10

$$For\ values\ from\ a\ population: \quad z = \frac{x - \mu}{\sigma}$$

$$z_{John} = \frac{2.85 - 3.0}{0.7} = -0.21$$

$$z_{Ali} = \frac{77 - 80}{10} = -0.3$$

John has the better GPA when compared to his school because his GPA is 0.21 standard deviations below his school's mean while Ali's GPA is 0.3 standard deviations below his school's mean.

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